



Namal University Mianwali

Department of Computer Science

1. COURSE DETAILS

Course Title:	Machine Learning
Course Code:	CS-341
Credits (Theory + Lab):	3(3+0)
Lecture Hours/Week:	3
Pre-Requisite:	Artificial Intelligence (CS-340)

Brief Course Description:

To develop intelligent software systems and information analysis from the pool of data, machine learning may provide key solutions. From simple decision making to an advanced automatic computer vision challenges in engineering and data science, machine learning is the primary approach these days. This course introduces the fundamental methods at the core of modern machine learning. Apart from theoretical foundation, it covers design and implementation of various supervised and unsupervised learning algorithms. The focus of this course is to introduce students to the fundamental and basic concepts of machine learning and deep learning and to develop skills that can enable them to use recent machine learning platforms for design and implementation purposes.

2. INSTURCTOR DETAILS

Name:	Dr. Khawar Khurshid
Lecture Timings:	As per Timetable
Office Hours:	Thursday 2:00pm – 5:00pm
Office Location:	CS Department
Office Phone:	Ext. 163
Email:	khawar.khurshid@namal.edu.pk

3. COURSE UNIT WISE MAJOR TOPICS

Unit No.	Topics	No. of Teaching Hours
1	Introduction	03
2	Unsupervised Learning	03
3	Supervised Learning	09
4	Bayesian Theory	06
5	Intro to Neural Networks	09
6	ANN, CNN, RNN	09
7	GANs, Encoder Decoder	06
Total Contact Hours		45

4. COURSE LEARNING OUTCOMES (CLOs)

Sr. No	Unit No	CLO Statement	Taxonomy Level	GA
CLO-1	1-7	Describe the fundamental concepts of machine learning in its broad categories.	C2	1
CLO-2	1-7	Apply different supervised and unsupervised algorithms to solve problems of various domains like Computer Vision, Signal Processing, Data Analysis.	C3	2
CLO-3	1-7	Design and implement machine learning algorithms using Python.	C5	4

5. WEEKLY LECTURE BREAKUP

Lecture #	Unit #	Topics Covered in Class	Reference in Book / Assessments
1	1	Introduction: What is machine learning?	Ch 01
2	1	Examples of machine learning applications Challenges and opportunities	Ch 01
3	1	Introduction to ML Environment: Installation and Usage of NumPy/SciKit/Matplotlib Jupyter Notebook	Online Resources
4	1	Data Processing in Python Introduction to Pandas	Online Resources
5	1	Unsupervised Learning: Clustering	Ch 07
6	1	K-means Fuzzy C-mean Applications of Clustering	Ch 07 Quiz 01
7	1	Supervised Learning: Feature spaces and Feature Vectors, Labels	Ch 08
8	1	Train/Test, Classification	Ch 09 Assignment 01
9	2	Bayesian Decision Theory: Probabilistic models and set theory	Ch 03
10	2	Bayes' Theorem Univariate classification	Ch 03
11	2	Bayesian Decision Theory: Multivariate problems	Ch 03
12	2	Independence and Correlation Naïve Bayes	Ch 03 Assignment 02
13	3	Dimensionality Reduction: PCA, LDA	Ch 06
14	3	SVMs	Ch 06 Quiz 02
15	3	Explatory Data Analysis: Feature extraction, weightage, visualization	Ch 06
16	3	SVD implementation Data Interpretation using PCA	Ch 06
Midterm Exam			
17	4	Linear Regression: Cost Function	Ch 04

			Project
18	4	Gradient Descent Challenges of local minima/maxima Overfitting	Ch 04
19	4	Logistic Regression: Model, Decision boundary	Ch 04
20	4	Loss function, Examples	Ch 04 Assignment 03
21	4	Introduction to Perceptron: Single Layer Perceptron	Ch 11
22	4	Basic Boolean functions	Ch 11 Quiz 03
23	4	Introduction to Neural Networks Introduction	Ch 12
24	4	Model Forward pass	Ch 12
25	5	Neural Networks: Back Propagation	Ch 12
26	5	Neural Network Weight calculations	Ch 12
27	5	Types of Neural Networks CNN	Ch 13
28	5	RNN, LSTM	Ch 13 Assignment 04
29	6	GANs	Online Resources
30	6	Encoders, Transformers	Online Resources Quiz 04
31	6	Project Presentations	
32	6	Project Presentations	
Final Term Exam			

6. CLO ASSESSMENT MECHANISM

Assessment Tool	CLO1	CLO2	CLO3		
Quizzes	✓	✓			
Assignments			✓		
Projects			✓		
Mid Term Exam	✓	✓			
Final Term Exam	✓	✓			

7. TEACHING MATERIAL

Textbook:

1. Introduction to Machine Learning by Ethem Alpaydin, Fourth Edition, 2020.

Reference Material:

1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow by Aurelien Geron, 2019.
2. Coursera ML course by Andrew Ng.

8. POLICY & PROCEDURES

- **Attendance Policy:** Students must attend 75% of this course's lectures. Students who fall short of the required attendance percentage cannot appear in the terminal examination.
- **Plagiarism Policy:** Namal University's rules and regulations prohibit plagiarism, copying, and other dishonest behavior. Violators will face serious disciplinary action.
- **Course Assessment:**

Assessment Tool	Quiz	Assignment	Project	Mid Term Exam	Final Term Exam	Total
Theory (T)	10%	10%	15%	25%	40%	100

9. VERIFICATION

(i) I verify that the content of this document is correct and up-to-date.

Dr. Khawar Khurshid Instructor's Name and Signature 	08-09-2025 Date
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(ii) I have reviewed course-outline and state that it complies with Namal University policies and guidelines.

Dr. Muhammad Ali Shahid Name and Signature of Head of Department	Date
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